



Hybrid ab initio and empirical machine learning models for solutions: learning the experimental ion pairing free energies

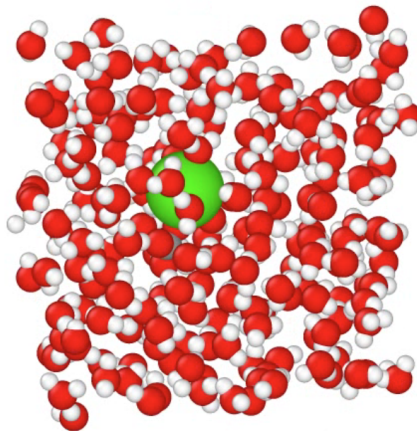
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Tutor: Pablo Piaggi

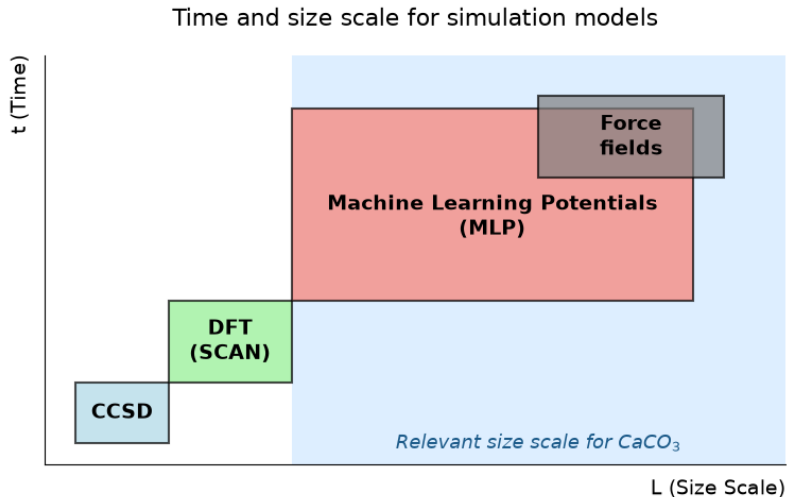
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Crystallization from solution

- Many essential materials are formed through crystallization from solutions.
- We chose calcium carbonate CaCO_3 our model system.
- Simulate the interactions in order to predict solid formation.
 - Quantum effects
 - Fitted to experimental data
 - Predictive power



Existing simulation methods



Ab initio Machine Learning Potentials (MLPs)

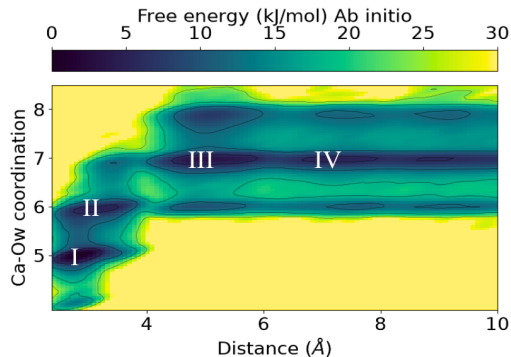
- **The Objective:** Achieve quantum (DFT) accuracy at a shorter computational cost, by training on a physically diverse, large-scale datasets of ab initio energies and forces.
- **Training:** learns to predict energies and forces using a large dataset.

$$\mathcal{L}_0(\theta) = p_E \sum_{i \in \mathcal{B}} (E(\mathbf{R}^i; \theta) - E_{ref}^i)^2 + p_f \sum_{i \in \mathcal{B}} \|\mathbf{f}(\mathbf{R}^i; \theta) - \mathbf{f}_{ref}^i\|^2$$

- **Physical Consistency:** predicts the forces: $\mathbf{f} = -\nabla_{\mathbb{R}} E$

Limitations of previous models

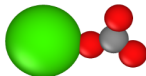
- Machine learning models capture the different states of ion pairing process.



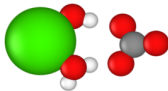
I) Bidentate



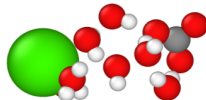
II) Monodentate



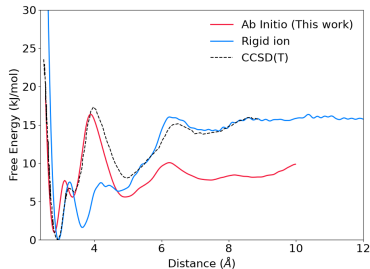
III) Solvent shared



IV) Solvent separated



- Machine learning models capture the different states of ion pairing process.
- Unable to capture thermodynamics

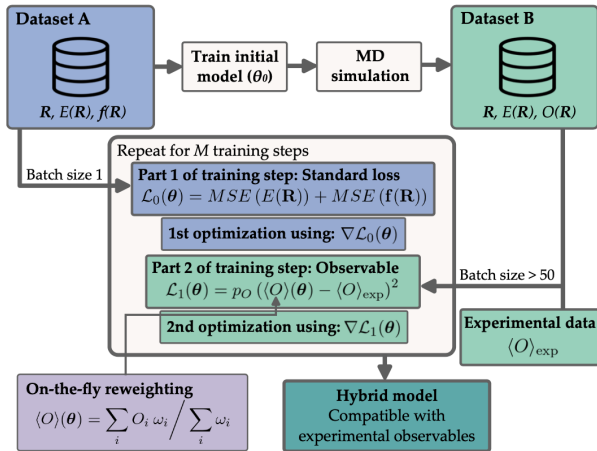


Sources: CCSD(T): Niamh O'Neill, Benjamin X. Shi, William C. Witt, et al., arXiv 2603.06800 (2026). SCAN-ML, Rigid ion and Polarizable: Pablo M. Piaggi, Julian D. Gale, and Paolo Raiteri, PNAS 122, e2415663122 (2025).

Solution of the problem

Train a machine learning model with ab initio and experimental data from an observable $\rightarrow$ ion pairing free energy ΔG .

Combining ab initio and experimental data



Source: Pablo Peña-Cano and Pablo M. Piaggi, arXiv 2511.14352 (2025).

Improving ion pairing free energy

$$\Delta G = -\frac{1}{\beta} \ln \left(\frac{\mathcal{P}(\text{Associated states})}{\mathcal{P}(\text{Non associated states})} \right)$$

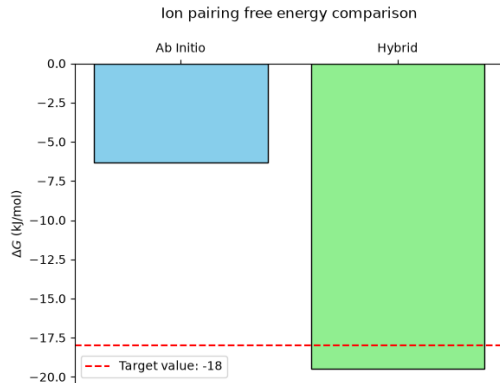
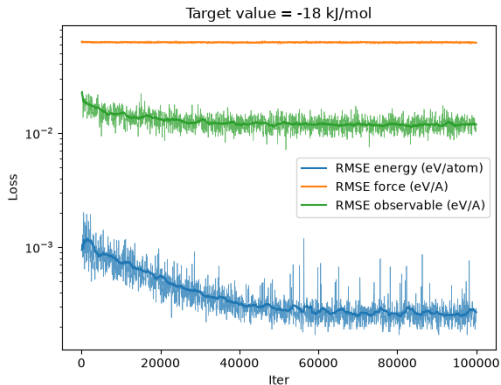
$$\Delta G = -\frac{1}{\beta} \ln \left(\frac{\sum_{r < r_1} e^{\beta V(r)}}{\sum_{r \in (r_1, r^*)} e^{\beta V(r)}} \right)$$

On-the-fly Reweighting

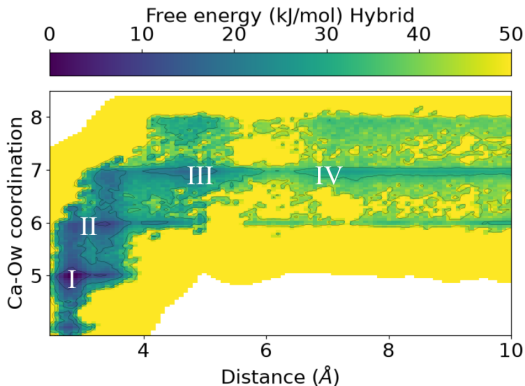
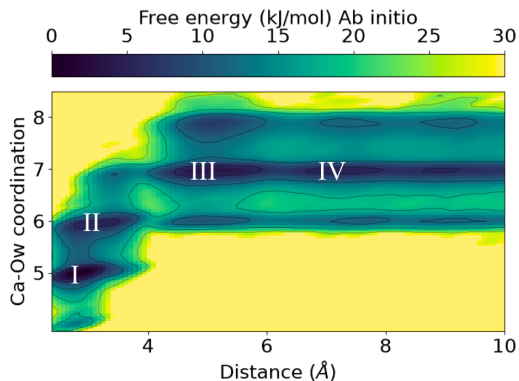
$$\Delta G = -\frac{1}{\beta} \ln \left(\frac{\sum_{r < r_1} e^{\beta V(r)}}{\sum_{r \in (r_1, r^*)} e^{\beta V(r)}} \right)$$

$$\Delta G(\theta) = -\frac{1}{\beta} \ln \left[C_0 V_{out} \cdot \left(\frac{\sum_{r < r_1} e^{\beta V(r)} e^{-\beta(E(r, \theta) - E_{ref}(r))}}{\sum_{r \in (r_1, r^*)} e^{\beta V(r)} e^{-\beta(E(r, \theta) - E_{ref}(r))}} \right) \right]$$

Success of the hybrid ML model



Consistency determining ion pairing steps



- **Bridging the gap:** Hybrid machine learning models successfully retain the quantum-level accuracy of *ab initio* calculations while performing precise predictions for macroscopic observables.
- **Thermodynamic applications:** This proves to be a powerful and efficient tool for studying the thermodynamic properties of complex solutions.
- **Future work paths:** Extending this framework to simulate larger-scale nucleation processes, exploring different solution concentrations, and applying the methodology to entirely new complex ionic systems.



Thank you for your attention.

Questions are welcomed!